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Hydrolyzed peptides from purple perilla (*Perilla frutescens* L. Britt.) seeds improve muscle synthesis and exercise performance in mice

Yixiang Liu[1] | Donghui Li[1] | Ying Wei[2] | Yu Ma[1] | Yuchen Wang[2] | Ling Huang[1] | Yanbo Wang[3]

[1]College of Food and Biological Engineering, Jimei University, Xiamen, People's Republic of China

[2]The Department of Food Engineering, China National Research Institute of Food & Fermentation Industries Corporation Limited, Beijing, People's Republic of China

[3]School of Food Science and Biotechnology, Zhejiang Gongshang University, Hangzhou, China

Correspondence
Ying Wei, The Department of Food Engineering, China National Research Institute of Food & Fermentation Industries Corporation Limited, Beijing, People's Republic of China.
Email: proudwy@126.com

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Abstract

The purple perilla (*Perilla frutescens* L. Britt.) seed peptides (PPSP) were obtained and their improvement of muscle synthesis and exercise performance was investigated in this work. Results showed that the weight-average molecular weight of the PPSP was 869 Dalton. The PPSP were rich in branched-chain amino acids (18.82 g/100 g) and anti-fatigue amino acids, including glutamate (Glu), aspartic acid (Asp), and arginine (Arg). After the administration of PPSP at 1.2 g kg^{-1} day^{-1} for 4 weeks, the muscle coefficient and muscle fiber thickness in mice displayed a distinct ($p < .05$) increase via the upregulation of myogenic differentiation (MyoD) and myogenin (MyoG). The improved muscle strength and exercise tolerance were also observed. Simultaneously, the levels of the biochemical blood markers associated with fatigue and the glycogen degradation in liver and muscle were significantly ($p < .05$) suppressed. These results suggested that PPSP could effectively promote muscle synthesis and ameliorate exercise fatigue.

Practical applications

Purple perilla is an annual herbal plant and widely grown in Asian countries as an important crop and food. It is believed that the protein content of purple perilla seeds can reach 23.7%, and the protein is rich in essential amino acids. However, the information about the beneficial effects of their proteins or peptides on muscle synthesis and anti-exercise fatigue were still limited. The present results discovered that the PPSP can effectively promote the growth of muscle tissue and improve exercise tolerance. It is indicated that PPSP may have a potential application value in partly or completely replacing animal proteins such as whey protein.

KEYWORDS

exercise fatigue, muscle growth, myogenic factor, peptides, purple perilla

Abbreviations: Arg, arginine; Asp, aspartic acid; BLA, blood lactic acid; BUN, blood urea nitrogen; Glu, glutamate; LDH, lactic dehydrogenase; LG, hepatic glycogen; MG, muscle glycogen; MyHC, myosin heavy chain; MyoD, myogenic differentiation; MyoG, myogenin; PPSP, purple perilla seed peptides; TP, total protein.

1 | INTRODUCTION

Muscle mass is controlled by the balance between muscle synthesis and degradation, which plays an important role in energy metabolism and exercise (Aoyama et al., 2019). The quality of muscle is primarily influenced by nutritional status and physical activity (Cohen, Nathan, & Goldberg, 2015). Nevertheless, malnutrition or disease can cause a person's muscles to atrophy, leading to a decline in health (Cohen et al., 2015; Landi et al., 2019). The loss of skeletal muscle mass and strength, for example, can lead to the loss of athletic ability and increase the risk of developing a chronic metabolic disease (DeGroot, Res, VanLoon, Saris, & Cermak, 2012). Therefore, improving muscle mass and strength is of paramount importance for people's health and physical performance (Wang, Hsieh, Farrar, & Ivy, 2020). Additionally, increasing muscle performance and stamina without violating the rules are of particular interest to athletes (Ommati, Farshad, Jamshidzadeh, & Heidari, 2019).

An increasing body of research reveals that higher protein intake can promote muscle synthesis (van der Meij, Deutz, Rodriguez, & Engelen, 2019). During prolonged resistance-type exercise training, protein supplementation can enhance muscle protein synthesis, thus increasing muscle mass and strength (MacKenzie-Shalders, Byrne, Slater, & King, 2015). Furthermore, it has been reported that casein supplementation increased the lean body mass and mean maximal power output of trained male cyclists (McLeay, Crum, Stannard, Barnes, & Starck, 2017). It is believed that whey protein (WP) intake can improve the muscle mass and quality of strength-trained or muscle building athletes, as well as other people who focus on aerobic exercises, such as marathons and swimming (Hulmi, Lockwood, & Stout, 2010). WP can also improve physiological adaptation during aerobic exercise and resistance training (Huang et al., 2017). However, several studies deal with the beneficial effects of vegetable protein on muscle synthesis and anti-exercise fatigue.

Purple perilla (*Perilla frutescens* L. Britt.) is an annual herbal plant widely grown in Asian countries as an essential food crop (Lee et al., 2020). In China, purple perilla has been a traditional crop for more than 2,000 years (Yu et al., 2017). It is well documented that purple perilla seeds serve a variety of physiological functions that include anti-allergy, anti-inflammatory, anti-oxidation, anticancer, antibacterial, and antidepression properties (Yu et al., 2017; Zhao, Yao-Yue, Qin, & Guo, 2012). According to previous studies, the protein content of purple perilla seeds can reach 23.7%, which is rich in essential amino acids (Longvah & Deosthale, 1998). However, limited research is available regarding the nutritional function of the proteins derived from purple perilla seeds.

Moreover, it is well-known that compared with proteins, active peptides are easier to digest and absorb (Song, Lee, & Song, 2015). Therefore, this work investigates the effect of hydrolyzed PPSP on muscle synthesis and anti-exercise fatigue. According to previous studies, WP can significantly promote muscle protein synthesis (Kanda et al., 2016). Consequently, WP is selected as a positive control, while the hydrolyzed peptides are obtained via the enzymatic hydrolysis of purple perilla seed proteins, after which the amino acid composition and molecular weight distribution of the peptides are analyzed. After a 4-week oral administration of PPSP to mice, the muscle synthesis and its associated regulatory factors are examined. In addition, the effect of PPSP supplementation in improving exercise performance and physiological adaptation for vigorous aerobic exercise are discussed further. This study provides information regarding the utilization of purple perilla seeds to develop nutritional supplements and functional foods that can improve exercise ability.

2 | MATERIALS AND METHODS

2.1 | Chemicals

Perilla protein (total protein (TP) content ≥ 90%) and WP (TP content ≥ 90%) were purchased from the Heilongjiang Zhenai Biological Technology Co. Ltd (Heilongjiang, China). Neutral protease (≥0.5 units/mg protein), papain (≥10 units/mg protein), 20 amino acid standards, and five peptide standards were purchased from Sigma-Aldrich (Guangzhou, China). The commercial kits for determining blood urea nitrogen (BUN), blood lactic acid (BLA), hepatic glycogen (LG), muscle glycogen (MG), lactic dehydrogenase (LDH), CRE, and TP were bought from the Institute of Biological Engineering of Nanjing Jiancheng (Nanjing, China). The enzyme-linked immunosorbent assay (ELISA) kits to determine the MyoD and MyoG of the mice were purchased from the Shanghai Zylien Biotechnology Co., Ltd (Shanghai, China). Other analytical reagent-grade solvents were purchased from the China National Pharmaceutical Industry Corporation Ltd. (Shanghai, China).

2.2 | The preparation of the PPSP

The protein powder derived from purple perilla seeds and deionized water were mixed at a mass-to-volume ratio of 1:20 (m/v), after which the pH of the solution was adjusted to 6.5 using hydrochloric acid. The protein homogenate was heated to 45°C, and the mixed enzymes (a mass ratio of neutral protease/papain, 10:1) were added at a mass-to-volume ratio of 1:1,000 (m/v). Following continuous stirring at 300 r/min for 4 hr, the enzymes in the solution were inactivated to obtain the crude peptide solution that was centrifuged at 8,000 r/min for 10 min, after which the supernatant was filtered through a ceramic membrane with a 10 nm pore diameter. Finally, after the filtered fluid was subjected to vacuum concentration and lyophilization, the PPSP powder was obtained and stored at −18°C for later use.

2.3 | Analysis of the amino acid composition

The amino acid analysis was performed using a method described in a previous study (Zhu et al., 2010). The PPSP sample (150 mg) was subjected to acid hydrolysis with 5 ml of 6 mol/L of HCl aqueous solution in a nitrogen atmosphere at 110°C for 24 hr. The hydrolysate was washed into a 50 ml volumetric flask and made up to the mark with distilled water. The amino acids were subjected to high performance liquid chromatography (HPLC) analysis (Agilent 1100) after pre-column derivatization with o-phthaldialdehyde or 9-fluorenylmethyl chloroformate. The levels of methionine and cysteine were determined separately via oxidation products before hydrolysis in 6 mol/L of HCl aqueous solution. The amino acid composition was reported as g of amino acid per 100 g of protein.

2.4 | The determination of the PPSP molecular weight distribution

The molecular weight distribution of PPSP was evaluated according to a method described in an earlier study (Qin et al., 2020), using an LC-20A HPLC system (Shimadzu, Kyoto, Japan) equipped with a TSK gel G2000 SWXL 300 × 7.8 mm column (Tosoh, Tokyo, Japan). The mobile phase consisted of acetonitrile/water (45:55, v/v) containing 0.1% (v/v) trifluoroacetic acid. Samples were eluted at a flow rate of 0.5 ml/min and monitored at 220 nm at 30°C. Tripeptide glycine (Gly)-Gly-Gly (189 Dalton), tetrapeptide Gly-Gly-Tyrosine (Tyr)-Arg (451 Dalton), bacitracin (1450 Dalton), aprotinin (6500 Dalton), and cytochrome C (12500 Dalton) were used as molecular weight standards.

2.5 | Animals and treatment

All animal procedures were performed following the Guidelines for Care and Use of Laboratory Animals of Jimei University (SCXK 2012-0005), and experiments were approved by the Animal Ethics Committee of China. Specific pathogen-free (SPF) male ICR mice (15 ± 2 g) were purchased from the Shanghai SLAC Laboratory Animal Co. Ltd (Shanghai, China). All mice were housed in individual stainless-steel cages and exposed to a 12-hr dark/light cycle in a clean SPF room. The temperature ranged from 22°C to 25°C, and the humidity was between 55% and 60%. The mice received a pellet diet (Shanghai SLAC Laboratory Animal Co. Ltd) and had free access to water. After an acclimatization period of 1 week, the mice were given a daily dose of PPSP. Then, 75 mice were randomly divided into five groups (15 mice per group): Group 1 was the control group (BL) (vehicle administration), Groups 2, 3, and 4 were the low-dose (LP) group (0.4 g kg^{-1} day^{-1}), medium-dose (MP) group (0.8 g kg^{-1} day^{-1}), and high-dose (HP) group (1.2 g kg^{-1} day^{-1}), respectively. Group 5 was administered with WP (1.2 g kg^{-1} day^{-1}) and was set as the positive control group. The mice received a prescribed dosage of PPSP or WP via intragastric administration for four consecutive weeks. All mice were weighed once a week.

2.6 | Viscera index analysis and muscle and adipose tissue collection

The viscera index was analyzed according to a method described in a previous study (Lin et al., 2015). The mice were sacrificed by cervical dislocation. The organs, such as the heart, liver, spleen, lungs, and kidneys were carefully removed, rinsed in saltwater, dried with filter paper, and weighed. The viscera index was expressed as the ratio of organ mass to mouse body weight. At the same time, the gastrocnemius and soleus muscles, as well as the epididymal, perirenal, and mesenteric adipose tissues, were also excised. Then, these muscles and adipose tissues were rinsed in saltwater, dried with filter paper, and weighed for utilization in subsequent experiments. The fat coefficient was calculated according to the ratio of fat weight to body weight. The muscle coefficient was expressed as the ratio of total muscle weight, including the soleus and gastrocnemius muscles, to body weight.

2.7 | Muscle strength test

The four limb hanging test was employed to evaluate the muscle strength according to a previous method (Bonetto, Andersson, & Waning, 2015). To ensure that the tested animals were adequately awake, the mice were brought to the experimental room 20 min before testing and were placed in the center of the wire mesh screen 40 cm above the padded surface. The four limb hanging time used to indicate muscular strength was recorded until the mice fell off.

2.8 | Myogenic factor investigation

Blood was collected after the mice were deprived of water and food for 12 hr, and the serum was prepared by centrifugation at 8,000 r/min for 5 min. Then, the myogenic factors such as MyoD and MyoG were analyzed using commercial kits.

2.9 | Histological analysis

The histological analysis of gastrocnemius muscles was carried out using a method described in previous studies (Botzenhart et al., 2020). After being immersed in paraffin, the gastrocnemius muscles were cut into 10 μm thick sections, after which the muscle tissues were stained with hematoxylin and eosin. The tissue morphology was observed with a light microscope (Olympus, Tokyo, Japan), while a minimum of four slices per muscle was analyzed. Histomorphometric measurements were performed on the images at 10× magnification using the image analysis software, Image J NIH, version 1.61 (https://imagej.nih.gov/ij/download.html). The total fiber area and fiber count were calculated with a minimum fiber area of 200 square microns, and these values were used to calculate the average area for each fiber.

2.10 | Exercise performance test

The exhaustive swimming test was conducted as previously described to evaluate the exercise performance of the mice (Wei et al., 2019). Briefly, after 30 min of the last gavage, the tails of the mice (five per group) were attached to a lead wire (5% of mouse body weight). The cube-shaped swimming pool (70 cm × 80 cm × 80 cm) was filled with warm water at a temperature of 30 ± 3°C. The endurance performance of each mouse was investigated according to the swimming time, which was recorded from the beginning to exhaustion. Exhaustion was confirmed when the mouse sank into the water and failed to swim to the water surface within 5 s.

2.11 | Physiological adaptation analysis

According to earlier studies, the LDH, BLA, CRE, and BUN levels in the blood were selected as the fatigue-associated biochemical indexes (Hu, Du, Du, Luo, & Xiong, 2020). After the exhaustive swimming test described in Section 2.10, the mice were removed from the water and dried with a hairdryer. Blood was collected after the mice had rested for 30 min. Then, 50 μl whole blood was mixed with 300 μl of the protein precipitator and centrifuged at 8,000 r/min for 5 min. The supernatant was collected and used to analyze BLA. For LDH, CRE, and BUN analysis, the blood samples were directly centrifuged at 8,000 r/min for 5 min, after which the supernatants were collected. The biochemical indexes mentioned above were measured using commercial kits. In addition, the gastrocnemius muscle and liver were used to determine the levels of MG and LG, respectively, according to a method described in a previous study (Hsu et al., 2016).

2.12 | Statistical analysis

All the experimental values were expressed as the mean ± standard deviation (*SD*). Differences between groups were analyzed with one-way analysis of variance (ANOVA), followed by Tukey tests. Results were considered statistically significant when $p < .05$. All the statistical analyses were performed with IBM SPSS Statistics 24.0.

3 | RESULTS

3.1 | The molecular weight distribution of PPSP

The molecular weight distribution of PPSP is shown in Table 1, indicating that the molecular weight range of PPSP was mainly below 2000 Dalton, where the 1,000–2,000 Dalton peptides accounted for 11.22 ± 1.96%, the 150–1,000 Dalton peptides accounted for 66.21 ± 4.25% and those <150 Dalton accounted for 11.92 ± 1.04%. In particular, peptides with a molecular weight of <1,000 Dalton accounted for more than 78%. In addition, the weight-average molecular weight of the obtained PPSP mixture was 869 Dalton.

3.2 | The amino acid composition of PPSP

Since previous studies have indicated that the rate of muscle synthesis is markedly dependent on the availability and abundance of amino acids (Bechshoeft et al., 2013), the amino acid composition of PPSP was investigated in this work. As shown in Table 2, 17 amino acids were discovered in PPSP, including seven amino acids essential for adults. Glu, Arg, Asp, and leucine (Leu) were the four most abundant amino acids in PPSP, accounting for 15.84 g/100 g, 10.81 g/100 g, 9.88 g/100 g, and 8.31 g/100 g, respectively. In addition, it is believed that proteins rich in branched amino acids are more conducive to muscle synthesis (Bechshoeft et al., 2013). According to the results shown in Table 2, PPSP also contains high levels of branched amino acids, including Leu, valine (Val), and isoleucine (Ile). The total branched amino acid in the PPSP was 18.82 g/100 g.

TABLE 1 The molecular weight distribution of PPSP

Molecular weight range (Dalton)	Weight-average molecular weight (Dalton)	Peak area percentage (%)
>10,000	11,231 ± 234	0.11 ± 0.02
5,000–10,000	6,577 ± 159	2.56 ± 0.13
3,000–5,000	3,829 ± 87	3.52 ± 0.15
2,000–3,000	2,437 ± 63	4.46 ± 0.21
1,000–2,000	1,404 ± 29	11.22 ± 1.96
150–1,000	421 ± 11	66.21 ± 4.25
<150	70 ± 5	11.92 ± 1.04

TABLE 2 The amino acid composition of PPSP

Items	Names of amino acid	Content (g/100 g)
1	Asp (Aspartic acid)	9.88 ± 0.06
2	Thr (Threonine)	3.03 ± 0.04
3	Ser (Serine)	4.00 ± 0.31
4	Glu (Glutamate)	15.84 ± 0.89
5	Gly (Glycine)	4.88 ± 0.11
6	Ala (Alanine)	3.87 ± 0.11
7	Val (Valine)	5.65 ± 0.22
8	Cys (Cysteine)	2.70 ± 0.32
9	Met (Methionine)	3.17 ± 0.06
10	Ile (Isoleucine)	4.86 ± 0.15
11	Leu (Leucine)	8.31 ± 0.21
12	Tyr (Tyrosine)	3.09 ± 0.32
13	Phe (Phenylalanine)	4.18 ± 0.04
14	His (Histidine)	2.32 ± 0.18
15	Lys (Lysine)	3.12 ± 0.10
16	Arg (Arginine)	10.81 ± 0.29
17	Pro (Proline)	3.30 ± 0.11

3.3 | The effect of PPSP on the body weight and viscera index of the mice

The effect of PPSP on the bodyweight of the mice was monitored during the 4-week administration period, and the results are shown

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in Figure 1. No significant ($p < .05$) differences in body weight were observed in any of the groups. The viscera index changes were also examined after the 4-week administration of PPSP, and the results are shown in Table 3. Compared with BL, no significant ($p < .05$) differences were discovered in either the WP group or the three PPSP groups.

3.4 | The effect of PPSP on exercise performance and muscle strength

The effect of PPSP on the exercise performance of the mice was determined using the exhaustive swimming test. Figure 2a shows that compared with the 75.1 min duration of the BL, the exhaustive swimming time of the WP and HP groups reached 105.4 min and 116.3 min, respectively. However, compared with the BL group, the LP and MP groups displayed better swimming performances, but no significant ($p < .05$) differences were evident. Figure 2b displays the effect of PPSP on the muscle strength of mice. The average limb hanging time in the BL group was only about 61 s. However, when the mice were administered with WP and PPSP for 4 weeks, the average limb hanging time increased to over 100 s. Therefore, it is clear that both WP and PPSP can improve the muscle strength of mice.

3.5 | The effect of PPSP on physiological adaptation during aerobic exercise

Following the exhaustive swimming test, the physiological adaptation of the mice was investigated. The LDH and BLA changes in the serum are shown in Figure 3a,b. Compared with the BL group, the administration of WP (1.2 g kg^{-1} day^{-1}) and PPSP (0.8 or 1.2 g kg^{-1} day^{-1}) can significantly ($p < .05$) increase the LDH level, resulting in a decline in BLA. Interestingly, the BLA level in the WP group was approximately the same as that in MP and HP groups. The concentrations of BUN and CRE are shown in Figure 3c,d, indicating that WP or PPSP supplementation can effectively suppress an increase in the BUN and CRE levels in mouse serum, while the PPSP was dose-dependent. Compared with the BL group, the levels of BUN and CRE in the MP group decreased from 20.45 mmol/L and 31.84 μmol/L to 17.21 mmol/L and 24.32 μmol/L, respectively. When the dose of PPSP was increased to 1.2 g kg^{-1} day^{-1}, the levels of BUN and CRE were further reduced to16.34 mmol/L and 15.16 μmol/L, respectively. Moreover, it is worth noting that the level of BUN in the HP group was markedly ($p < .05$) lower than in the WP group. The changes in the LG and MG levels in the mice are presented in Figure 3e,f. Compared with the BL group, a significant ($p < .05$) increase was observed in the LG and MG levels when the mice were administered with WP and PPSP at a dose of 1.2 g kg^{-1} day^{-1}. As depicted in Figure 3e, the LG level in the HP group was substantially higher ($p < .05$) than that in the WP group.

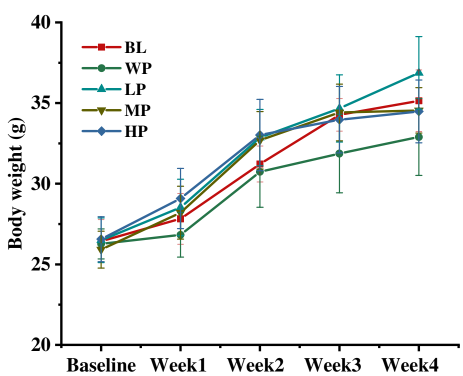

FIGURE 1 The effect of PPSP on the bodyweight of mice. The BL mice were administered with distilled water as a vehicle. WP was the positive control group, and the dosage was 1.2 g kg^{-1} day^{-1}. LP, MP, and HP doses were 0.4 g kg^{-1} day^{-1}, 0.8 g kg^{-1} day^{-1}, and 1.2 g kg^{-1} day^{-1}, respectively. Data are expressed as mean ± *SD* ($n = 5$)

3.6 | The effect of PPSP on the fat and muscle coefficients

After the mice were administered with WP and PPSP for 4 weeks, the fat and muscle coefficients were analyzed. Figure 4 shows that the intake of PPSP efficiently reduced the fat coefficient and increased the muscle coefficient of the mice. Compared with the BL group, the fat coefficient in the HP group reduced from 38.2 mg/g to 32.4 mg/g, while the muscle coefficient increased from 9.8 mg/g to 11.3 mg/g. Furthermore, WP also improved the muscle coefficient but had a slight inhibitory effect on the fat coefficient.

3.7 | The effect of PPSP on the MyoD, MyoG, and TP in the serum

Previous studies have indicated that muscle synthesis is regulated by myogenic regulators such as MyoG and MyoD (Xu et al., 2018). In addition, the total serum protein, including albumin and globulin is an important index for the evaluation of malnutrition (Paiva et al., 2012; Park & Kim, 2019). Therefore, the levels of MyoG, MyoD, and TP in the serum were analyzed after a 4-week PPSP administration. Figure 5 indicates that compared with the BL group, the levels of MyoG, MyoD, and TP were significantly ($p < .05$) increased when the mice were fed WP and PPSP. Interestingly, the MyoD serum level in the HP group was higher ($p < .05$) than in the WP group.

3.8 | The histopathological observation of the muscle tissue

The morphology and distribution of the muscle fibers are displayed in Figure 6, indicating that no significant difference was evident in the shape of the muscle fibers. According to the quantification

TABLE 3 The effect of PPSP on the viscera index

Groups	Heart index (mg/g)	Liver index (mg/g)	Spleen index (mg/g)	Lung index (mg/g)	Kidney index (mg/g)
BL	5.18 ± 0.49	58.23 ± 2.64	2.69 ± 0.55	6.9 ± 1.19	19.68 ± 0.26
WP	5.60 ± 0.31	55.3 ± 3.21	2.55 ± 0.35	7.24 ± 1.02	19.35 ± 1.71
LP	5.32 ± 0.19	54.97 ± 4.27	3.17 ± 0.38	7.08 ± 2.71	17.81 ± 1.19
MP	5.10 ± 0.64	51.54 ± 2.62	3.41 ± 0.49	5.96 ± 0.17	18.48 ± 0.95
HP	5.45 ± 0.32	54.46 ± 3.44	2.79 ± 0.53	5.65 ± 0.46	17.53 ± 1.82

Note: The BL mice were administered with distilled water as a vehicle. WP was the positive control group, and the dosage was 1.2 g kg^{-1} day^{-1}. LP, MP, and HP doses were 0.4 g kg^{-1} day^{-1}, 0.8 g kg^{-1} day^{-1}, and 1.2 g kg^{-1} day^{-1}, respectively. Data are expressed as mean ± *SD* ($n = 5$).

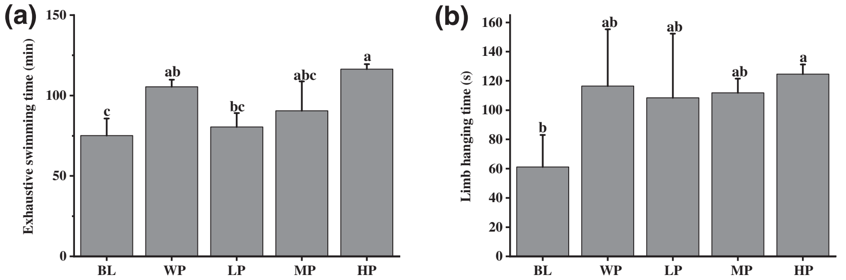

FIGURE 2 The effect of PPSP on exhaustive swimming time (a) and four limb hanging time (b). The BL mice were administered with distilled water as a vehicle. WP was the positive control group, and the dosage was 1.2 g kg^{-1} day^{-1}. The LP, MP, and HP doses were 0.4 g kg^{-1} day^{-1}, 0.8 g kg^{-1} day^{-1}, 1.2 g kg^{-1} day^{-1}) respectively. Data are expressed as mean ± *SD* ($n = 5$). Different letters indicate significant differences ($p < .05$)

data of the same size image (3,000 × 2,200 pixels), the PPSP was dose-dependent, promoting the thickening of the muscle fibers. Compared with the BL group (1,228.48 ± 27.00 μm^2), a significantly ($p < .05$) larger cross-sectional area of muscle fibers was evident in the MP (1,419.15 ± 156.62 μm^2) and HP (1,736.17 ± 84.90 μm^2) groups. Additionally, no significant ($p < .05$) differences were apparent between the muscle fiber thickness of the HP and WP groups.

4 | DISCUSSION

Several studies have shown that compared with proteins, peptides are more conducive to absorption by intestinal enterocytes and is reportedly more helpful in extensive exercise (Yu, Lu, Bie, Lu, & Huang, 2008). It is believed that supplementation with grass carp peptides was better than protein in improving the swimming endurance of mice (Ren, Zhao, Wang, Cui, & You, 2011). According to previous studies, the peptides from the *Hippocampus abdominalis* could protect muscle fibers and prolong the running time of mice (Zhang et al., 2019), and loach peptides prepared by papain digestion could increase the endurance of mice by activating the energy metabolism (You, Zhao, Regenstein, & Ren, 2011), while soybean peptides produced by fermentation could enhance the exercise ability and alleviate physical fatigue in mice (Yu et al., 2008). Therefore, PPSP was prepared, and its enhancing effect on the muscle synthesis and exercise performance of mice was evaluated.

The promotional effect of dietary peptides or protein on muscle synthesis is closely related to its amino acid composition (Zdzieblik, Oesser, Baumstark, Gollhofer, & König, 2015). It is believed that the proteins rich in branched-chain amino acids, including Leu, Val, and Ile, are more beneficial to muscle growth (Torre-Villalvazo, Alemán-Escondrillas, Valle-Ríos, & Noriega, 2019), since these amino acids are involved in regulating the metabolic pathway of muscle protein synthesis, which is crucial for improving athletic performance (Campos-Ferraz, Bozza, Nicastro, & Lancha, 2013). For example, supplementation with branched-chain amino acids can efficiently reduce serum LDH levels and muscle damage during the endurance exercise of soccer players (Shekarchizadeh, Enayat, & Karimian, 2018). It is also believed that the Leu can improve exercise ability by reducing the degradation of glycogen and the catabolism of muscle protein (Zhang et al., 2017). In this study, the level of the branched-chain amino acids in the PPSP was as high as 18.82 g/100 g, while previous studies indicated that the content of these amino acids in WP is about 22.78 g/100 g (Bechshoeft et al., 2013). Therefore, this study showed that both PPSP and WP improved muscle synthesis.

However, PPSP performed better than WP in promoting physiological adaptation during the high-intensity aerobic exercise in this study, such as suppressing the serum BUN level and reducing the degradation of LG. This may be related to the fact that PPSP is also rich in amino acids with anti-exercise fatigue properties, such as Arg, Glu, and Asp. The content of these three amino acids in PPSP reached a level of up to 36.5 g/100 g. It is believed that Arg supplementation can reduce the oxygen consumption of

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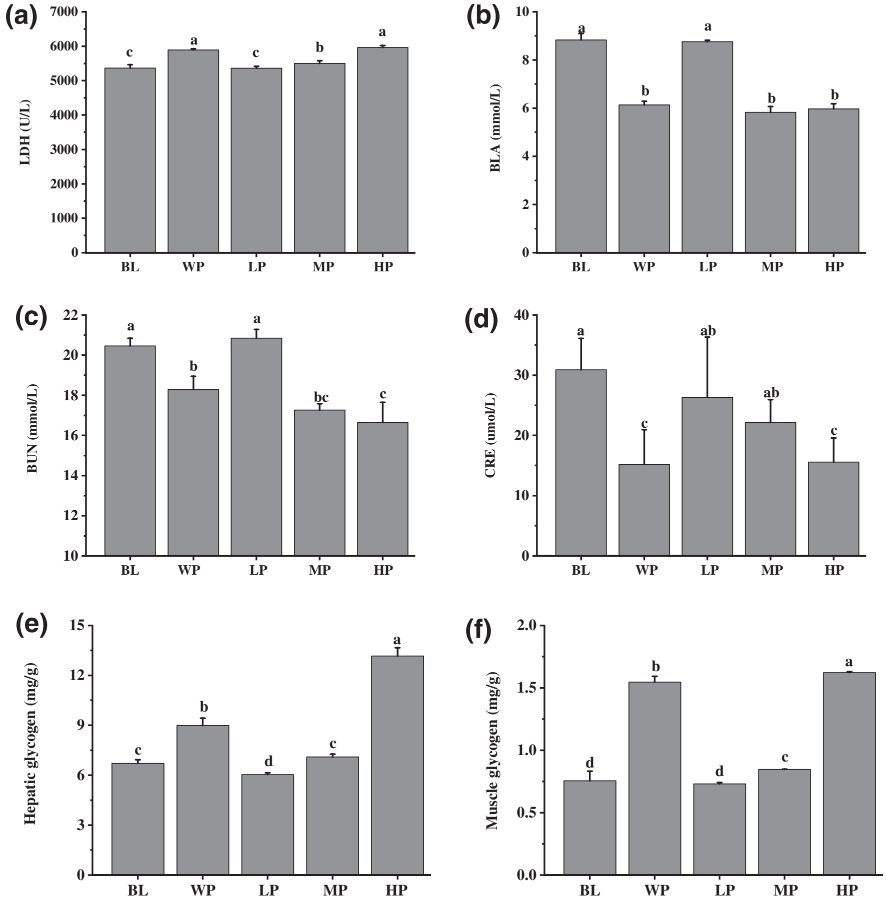

FIGURE 3 The effect of PPSP on LDH (a), BLA (b), BUN (c), CRE (d), LG (e), and MG (f). The BL mice were administered with distilled water as a vehicle. WP was the positive control group, and the dosage was 1.2 g kg^{-1} day^{-1}. LP, MP, and HP doses were 0.4 g kg^{-1} day^{-1}, 0.8 g kg^{-1} day^{-1}, and 1.2 g kg^{-1} day^{-1}, respectively. Data are expressed as mean ± *SD* ($n = 5$). Different letters indicate significant differences ($p < .05$)

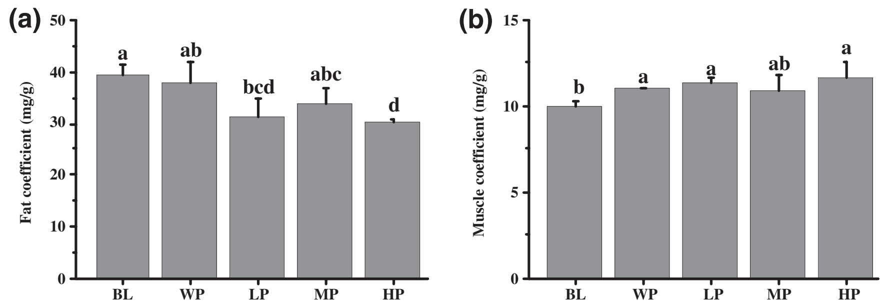

FIGURE 4 The effect of PPSP on the fat (a) and muscle (b) coefficients. The BL mice were administered with distilled water as a vehicle. WP was the positive control group, and the dosage was 1.2 g kg^{-1} day^{-1}. The LP, MP, and HP doses were 0.4 g kg^{-1} day^{-1}, 0.8 g kg^{-1} day^{-1}, and 1.2 g kg^{-1} day^{-1}, respectively. Data are expressed as mean ± *SD* ($n = 5$). Different letters indicate significant differences ($p < .05$)

moderate-intensity exercise and reduce the elevation of blood ammonia and lactic acid caused by exercise (Bailey et al., 2010). Previous studies have also shown that the Arg can regulate fat deposition, promote the slow expression of myosin heavy chain (MyHC) in porcine skeletal muscle satellite cells, and promote the transformation of the muscle fiber type from fast-twitch to slow-twitch (Chen et al., 2018). Asp was discovered to be helpful in oxidative deamination and reducing the concentration of blood ammonia, resulting in the delay of fatigue occurrence (Ding et al., 2011). Furthermore, Glu is thought to have a positive effect on the nervous system while also helping to reduce mental and physical fatigue (Sun et al., 2014).

Muscle growth is the result of hyperplasia and hypertrophy of muscle fibers (Rescan, 1998). In mammalian skeletal muscles, the increase in the number of muscle fibers stops shortly after birth, and further muscle growth is primarily driven by hypertrophy (the growth of existing fibers) (Sun et al., 2020). It is believed that the growth of muscle fibers, namely myogenesis, is regulated by myogenic

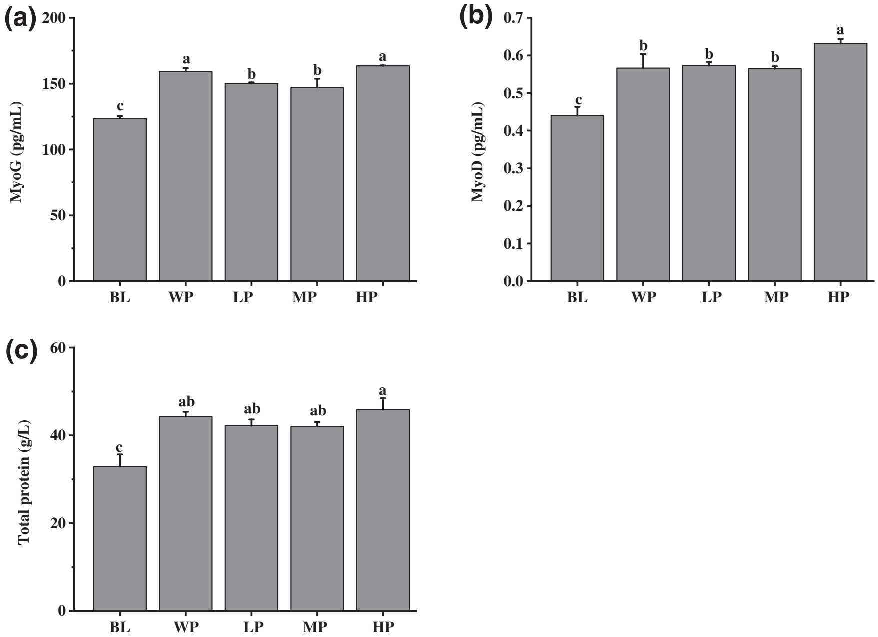

FIGURE 5 The effect of PPSP on the MyoG (a), the MyoD (b), and the TP (c) in the serum. The BL mice were administered with distilled water as a vehicle. WP was the positive control group, and the dosage was 1.2 g kg^{-1} day^{-1}. The LP, MP, and HP doses were 0.4 g kg^{-1} day^{-1}, 0.8 g kg^{-1} day^{-1}, and 1.2 g kg^{-1} day^{-1}, respectively. Data are expressed as mean ± *SD* ($n = 5$). Different letters indicate significant differences ($p < .05$)

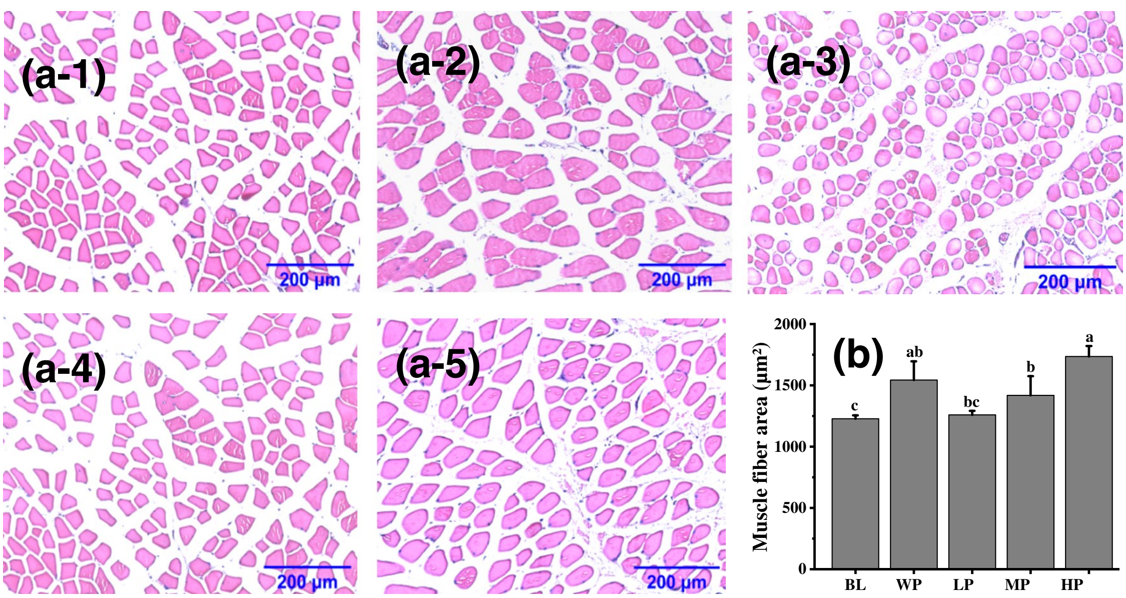

FIGURE 6 The histopathological observation of the gastrocnemius muscle. (a-1) to (a-5) represents the muscle fiber morphology of the BL, WP, LP, MP, and HP groups, respectively. (b) The cross-sectional area of muscle fiber. The BL mice were administered with distilled water as a vehicle. WP was the positive control group, and the dosage was 1.2 g kg^{-1} day^{-1}. The LP, MP, and HP doses were 0.4 g kg^{-1} day^{-1}, 0.8 g kg^{-1} day^{-1} and 1.2 g kg^{-1} day^{-1}, respectively. Data are expressed as mean ± *SD* ($n = 5$). Different letters indicate significant differences ($p < .05$)

regulatory factors (Frias et al., 2016). MyoD is the main regulator of muscle growth and differentiation and is essential for muscle regeneration by regulating the muscle fiber type (Xu et al., 2018). MyoG is another key regulator of myogenesis and is necessary for skeletal muscle development and myoblast differentiation during myogenesis (Moresi et al., 2010). In previous studies, the levels of MyoD and MyoG proteins in the blood were consistent with the MyoD and MyoG mRNA expression levels (Xu et al., 2018). Therefore, this study analyzed the changes in the MyoD and MyoG proteins in the blood, indicating that the levels of both were enhanced after the mice were administered with PPSP or WP. Additionally, the plasma concentration of TP is reportedly related to the transformation of the muscle

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fiber type (Paiva et al., 2012). This study demonstrates that PPSP intake can increase the TP content in mouse blood. The four limb hanging test, also known as the Kondziella's inverted screen test, is one of the most common methods of assessing the strength of laboratory mice (Bonetto et al., 2015). A reduced tendency to fall in the four limb hanging test indicated an improvement in muscle strength (Bárez-López et al., 2014). According to a previous study, both WP supplementation and exercise training could improve muscle strength (Chen, Huang, Chiu, Chang, & Huang, 2014). In the present study, a longer limb hanging time was observed after the mice were treated with PPSP for 4 weeks, indicating that PPSP intake can increase muscle strength. This hypothesis was further confirmed by the histopathological observation in this work, in which coarsened muscle fibers were evident after PPSP treatment.

The improvement of the forced swimming time is an important index reflecting the anti-fatigue function of the active substances. The mice administered with PPSP at a dose of 1.2 g kg^{-1} day^{-1} displayed a significantly longer exhaustive swimming time than the BL. This result revealed that PPSP improved exercise endurance and reduced physiological fatigue. Furthermore, to demonstrate the fatigue resistance of PPSP, several biochemical indexes related to fatigue were determined. Since LDH is responsible for removing BLA, it can reflect the degree of lactate metabolism (Zhang et al., 2017). BLA is an important index of glucose anaerobic metabolism, which is the final product of glycolysis during high-intensity exercise (Yang et al., 2020). The high levels of BLA produced by aerobic exercises, such as intensive swimming can reduce the pH value in muscles and blood, damage certain organs, and eventually lead to fatigue (Luo et al., 2019). BUN is another representative index of fatigue, and its concentration in the blood can reflect the metabolism of protein and amino acids (Zhang et al., 2020). Consequently, high BUN levels in the blood indicate an increase in protein decomposition as well as the occurrence of fatigue (Zhang et al., 2020). In addition, CRE is the final product of creatine and creatine phosphate metabolism, which is produced by the spontaneous non-enzymatic dehydration of CRE in muscle cells (Kashani, Rosner, & Ostermann, 2020). Both LG and MG are directly related to physical activity and endurance (Tan et al., 2012). When the body is engaged in vigorous exercise, glycogen is first utilized to provide energy (Tan et al., 2012). Therefore, increased storage of LG and MG contributes to higher endurance and exercise capacity. Compared with the BL, the BLA, BUN, and CRE levels in the blood were significantly ($p < .05$) reduced when the mice were administered with PPSP at doses of 0.8 g kg^{-1} day^{-1} or 1.2 g kg^{-1} day^{-1}. Moreover, a significant ($p < .05$) increase was observed in the serum LDH activity, while both the LG and MG levels increase remarkably ($p < .05$) after PPSP intake. Therefore, it is evident that supplementation with PPSP is beneficial for improving exercise performance.

5 | CONCLUSIONS

This study obtains hydrolyzed peptides from purple perilla seeds and investigates their enhancing effect on muscle synthesis and exercise performance. The weight-average molecular weight of PPSP is 869 Dalton, of which peptides with a molecular weight <1,000 accounted for about 78%. The PPSP does not only display an abundance of branched-chain amino acids that promote muscle growth but it is also rich in amino acids that ameliorate exercise fatigue, such as Arg, Glu, and Asp. After the administration of PPSP for 4 weeks, the muscle coefficient and muscle fiber thickness in the mice exhibit a distinct increase via the upregulation of myogenic regulators, such as MyoG and MyoD. Furthermore, improved muscle strength is observed in the mice was after the supplementation with PPSP. In addition, PPSP intake can effectively decrease the BLA, BUN, and CRE levels in the blood, increasing the LG and MG, and improving the exercise endurance of the mice. Therefore, PPSP has the potential to be developed into functional foods for enhancing muscles and preventing exercise fatigue.

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AUTHOR CONTRIBUTIONS

Yixiang Liu: project Administration, writing Original Draft; Donghui Li: data Curation, formal Analysis, investigation, methodology; Ying Wei: conceptualization, funding Acquisition; Yu Ma: investigation, resources, software; Yuchen Wang: funding Acquisition, validation; Ling Huang: supervision, visualization; Yanbo Wang: writing Review Editing.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

ORCID

Yixiang Liu https://orcid.org/0000-0002-4574-8309

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